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**Final Report**

Date: 31.01.2018

Patient name: **Al Otaibi, Eleen Talal**Your ref.: **MRN: 27011592**DOB (dd.mm.yyyy): **14.03.2016**Sample collection date (dd.mm.yyyy): **11.12.2017**Order received (dd.mm.yyyy): **03.01.2018**Sex: **female**Patient no.: **1256164**Sample type: **blood, filter card**Order no.: **62437132**

**Whole Exome Sequencing (CentoXome GOLD®) and Mitochondrial Genome Sequencing (CentoMito® Genome) were performed.**

**Clinical information:** 20 months old female with frequent visits to ER due to hyperactive airway disease, after 6 months of age she started to have progressive loss of milestones with ataxia, nystagmus, ptosis, eye squint and hyperreflexia with clonus, upper motor neuron lesions (UMNL), normal ammonia-lactate; age of manifestation at 14 months. Family history: positive for epilepsy and early deaths (according to the provided pedigree two nephews deceased at age of 6 ½ years and 1 year, clinically diagnosed with progressive delayed epilepsy); the parents are consanguineous, no affected siblings

**Variants with possible relevance to patient's phenotype:****NAXE (NM\_144772.2, sequencing)****homozygous variant c.262G>T p.(Gly88Trp)**

Please see table below for a detailed description of the detected variants

**The genetic diagnosis of progressive, early-onset encephalopathy with brain edema and/or leukoencephalopathy is possible.**

**Interpretation**

**The NAXE variant c.262G>T p.(Gly88Trp) causes an amino acid change from Gly to Trp at position 88. It is classified as variant of uncertain significance (class 3) according to the recommendations of Centogene and ACMG (please, see additional information below).**

Pathogenic variant in the NAXE gene are associated with progressive, early-onset encephalopathy with brain edema and/or leukoencephalopathy, an autosomal recessive disorder (OMIM®: 61718). It is characterized by rapidly progressive neurologic deterioration that is usually associated with a febrile illness. Affected infants tend to show normal early development followed by acute psychomotor regression with ataxia, hypotonia, respiratory insufficiency and seizures, resulting in coma and death in the first years of life. Brain imaging shows multiple abnormalities, including brain edema and signal abnormalities in the cortical and subcortical regions (PMID: 27616477). The genotype-phenotype correlation has been first described by Spiegel et al., 2016 (PMID: 27122014), who examined 5 siblings of a consanguineous family presented first noted symptoms at 6 - 12 months: typically associated with febrile illnesses, progressive generalized weakness of predominant proximal distribution, milestones previous acquired were lost within days, gradual development of generalized encephalopathy and loss of cognitive skills. At 2 years of age the siblings became bedridden, respiratory muscle failed and mechanically ventilation was required. Four of the patients died between 1 and 3 years of age.

Additionally the homozygous DCHS1 variant c.391G>T p.(Val131Leu) has been detected, which causes an

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amino acid change from Val to Leu at position 131. So far it has not been published and is classified as variant of uncertain significance (class 3) according to the recommendations of Centogene and ACMG (please, see additional information below).

Pathogenic variants in the DCHS1 gene are associated with van Maldergem syndrome type 1, an autosomal recessive disorder (VMLDS1; OMIM®: 601390). The phenotype described for your patient shows a partial overlap with this disorder. VMLDS1 is characterized by intellectual disability, typical craniofacial features, auditory malformations resulting in hearing loss and skeletal and limb malformations. Some patients have renal hypoplasia. Brain MRI typically shows periventricular nodular heterotopia (PMID: 24056717).

In the mitochondrial genome no clinically relevant variant was detected by sequencing. Please note that pathogenic variants encoded in the mitochondrial genome may not be detected in samples derived from blood if the level of heteroplasmy is low (typically <15% heteroplasmy).

### Recommendations

We recommend parental carrier testing to confirm homozygosity by excluding the presence of a large deletion. We also recommend carrier testing for all informative family members to establish whether the variant is associated with the disorder or not. Genetic counselling is recommended.

### Incidental findings

We did not detect any class 1 or 2 variants in the genes for which incidental findings are reported based on the ACMG guidelines.

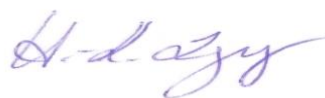
If you have any further questions or analysis requests, please do not hesitate to contact us at [support@centogene.com](mailto:support@centogene.com).

Best regards,



**Prof. Dr. Arndt Rolfs, MD**

Chief Medical Director



**Dr. Anne-Kathrin Langenberg,  
PhD**

Scientist

### Analysis statistics WES

| Average      | % target bp covered |         |         |         |         |         |
|--------------|---------------------|---------|---------|---------|---------|---------|
| coverage (X) | 0X                  | ≥ 1X    | ≥ 5X    | ≥ 10X   | ≥ 20X   | ≥ 50X   |
| 103.478      | 0.349247            | 99.6508 | 99.1668 | 98.0672 | 93.9943 | 71.8625 |

### Statistics (CentoMito Genome)

Overall, 100% of the target base pairs of the mitochondrial genes were covered at least 1000x. The average cover was: 33043,6x.

### CENTOGENE variant classification (based on ACMG recommendations)

- Class 1** – Pathogenic
- Class 2** – Likely pathogenic
- Class 3** – Variant of uncertain significance (VUS)
- Class 4** – Likely benign
- Class 5** – Benign
- Class 6** – Disease-associated variant

### Detailed description of the detected variants

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| Gene  | Variant Coordinates                                                 | In silico parameters*                                                                                                                          | Allele Frequencies**                           | Type and Classification***                      |
|-------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------|
| NAXE  | Chr1(GRCh37):g.156561972G>T<br>NM_144772.2:c.262G>T<br>p.(Gly88Trp) | PolyPhen: Probably damaging<br>Align-GVGD: -<br>SIFT: Deleterious<br>MutationTaster: Disease causing<br>Conservation: nt high                  | gnomAD: -<br>ESP: -<br>1000 G: -<br>CentoMD: - | Missense<br>Uncertain significance<br>(class 3) |
| DCHS1 | Chr11(GRCh37):g.6662454C>A<br>NM_003737.3:c.391G>T<br>p.(Val131Leu) | PolyPhen: Possibly damaging<br>Align-GVGD: C0<br>SIFT: Deleterious<br>MutationTaster: Disease causing<br>Conservation: nt moderate/aa moderate | gnomAD: -<br>ESP: -<br>1000 G: -<br>CentoMD: - | Missense<br>Uncertain significance<br>(class 3) |

Variant description based on Alamut Batch (latest database available). \* AlignGVGD: C0: least likely to interfere with function, C65: most likely to interfere with function, splice prediction tools: SSF, MaxEnt, HSF. \*\* Genome Aggregation Database (gnomAD), Exome Sequencing Project (ESP), 1000Genome project (1000G) and CentoMD® (latest database available). \*\*\* based on ACMG recommendations

## Methods

RNA capture baits against approximately 60 Mb of the Human Exome (targeting >99% of regions in CCDS, RefSeq and Gencode databases) is used to enrich regions of interest from fragmented genomic DNA with Agilent's SureSelect Human All Exon V6 kit. The generated library is sequenced on an Illumina platform to obtain an average coverage depth of ~100x. Typically, ~97% of the targeted bases are covered >10x. An end to end in-house bioinformatics pipeline including base calling, alignment of reads to GRCh37/hg19 genome assembly, primary filtering out of low quality reads and probable artefacts, and subsequent annotation of variants, is applied. All disease causing variants reported in HGMD®, in ClinVar or in CentoMD® as well as all variants with minor allele frequency (MAF) of less than 1% in gnomAD database are considered. Evaluation is focused on coding exons along with flanking +/-20 intronic bases. All pertinent inheritance patterns are considered. In addition, provided family history and clinical information are used to evaluate eventually identified variants. All identified variants are evaluated with respect to their pathogenicity and causality, and these are categorized into classes 1 - 6 (see above). All variants related to the phenotype of the patient, except benign or likely benign variants, are reported.

Variants of relevance identified by NGS are continuously and individually in-house validated for quality aspects; those variants which meet our internal QC criteria (based on extensive validation processes) are not validated by Sanger.

Targeted amplification of the entire mitochondrial genome is done using two overlapping long range PCRs. The amplicons are run on the Bioanalyzer to assess for any putative deletion within the mitochondrial genome. The amplified products are subsequently tagged and Illumina compatible adapters ligated to generate libraries that are sequenced on Illumina platforms to an average sequencing depth of 1000x or more.

Raw sequence data analysis, including base calling, demultiplexing, alignment to the Revised Cambridge Reference Sequence (rCRS) of the Human Mitochondrial DNA (NC\_012920) and variant calling are performed using validated in-house software. Following the base calling and primary filtering of low quality reads, standard Bioinformatics pipeline was implemented to annotate detected variants and to filter out probable artefacts. The pipeline confidently detects heteroplasmy levels down to 15%.

All identified variants are evaluated with respect to their pathogenicity and causality, and these are categorized into classes 1 - 6 (see above). All variants related to the phenotype of the patient, except benign or likely benign variants, are reported.

Variants of relevance identified by NGS are continuously and individually in-house validated for quality aspects; those variants which meet our internal QC criteria (based on extensive validation processes) are not validated by Sanger.

## Limitations

Test results are interpreted in the context of clinical findings, family history and other laboratory data. Only variations in genes potentially related to the proband's medical condition are reported. Rare polymorphisms may lead to false negative or positive results. Misinterpretation of results may occur if the information provided is inaccurate or incomplete. If results obtained do not match the clinical findings, additional testing should be considered.

Specific genetic events like copy number variants, translocations and repeat expansions may not be reliably detected with Exome Sequencing. In addition, due to limitations in technology, certain regions may either not be covered or may be poorly covered, where variants cannot be confidently detected.

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#### Additional information

This test was developed and its performance validated by CENTOGENE AG. The US Food and Drug Administration (FDA) has determined that clearance or approval of this method is not necessary and thus neither have been obtained. This test has been developed for clinical purposes. All test results are reviewed, interpreted and reported by our scientific and medical experts.

In line with ACMG recommendations for reporting of incidental findings in clinical exome and genome sequencing (Genetics in Medicine, 2016), we report incidental findings, i.e. pathogenic variants (class 1) and likely pathogenic variants (class 2) only in the recommended genes for the recommended phenotypes.

To also exclude mistaken identity in your clinic, several guidelines recommend testing a second sample that is independently obtained from the proband. Please note that any further analysis will result in additional costs.

The classification of variants can change over the time. Please feel free to contact CENTOGENE ([testing@centogene.com](mailto:testing@centogene.com)) in the future to determine if there have been any changes in classification of any reported variants.

#### Disclaimer

Any preparation and processing of a sample from patient material provided to CENTOGENE by a physician, clinical institute or a laboratory (by a "Partner") and the requested genetic and/or biochemical testing itself is based on the highest and most current scientific and analytical standards. However, in very few cases genetic or biochemical tests may not show the correct result, e.g. because of the quality of the material provided by a Partner to CENTOGENE or in cases where any test provided by CENTOGENE fails for unforeseeable or unknown reasons that cannot be influenced by CENTOGENE in advance. In such cases, CENTOGENE shall not be responsible and/or liable for the incomplete, potentially misleading or even wrong result of any testing if such issue could not be recognized by CENTOGENE in advance.

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